List of named distributions

$X \sim \text{Bernoulli}(p)$, PMF: $p_X(0) = 1 - p, \quad p_X(1) = p$

$X \sim \text{Binomial}(n, p)$, PMF: $p_X(k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad 0 \le k \le n$

$X \sim \text{Geometric}(p)$, PMF: $p_X(k) = p(1-p)^{k-1}, \quad k = 1, 2, \ldots$

$X \sim \text{Poisson}(\lambda)$, PMF: $p_X(k) = \frac{\lambda^k}{k!} e^{-\lambda}, \quad k = 0, 1, 2, \ldots$

$X \sim \text{Hypergeometric}(N, N_A, n)$, PMF: $p_X(k) = \frac{\binom{N_A}{k}\binom{N-N_A}{n-k}}{\binom{N}{n}}, \quad 0 \le k \le n$

$X \sim \text{Uniform}(a, b)$, PDF: $f_X(x) = \begin{cases} \frac{1}{b-a} & \text{for } x \in [a, b], \\ 0 & \text{otherwise} \end{cases}$

$X \sim \text{Normal}(\mu, \sigma^2)$, PDF: $f_X(t) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(t-\mu)^2}{2\sigma^2}}$

$X \sim \text{Exponential}(\lambda)$, PDF: $f_X(t) = \begin{cases} \lambda e^{-\lambda x} & \text{for } x > 0, \\ 0 & \text{otherwise} \end{cases}$

$X \sim \text{Gamma}(r, \lambda)$, PDF: $f_X(t) = \begin{cases} \frac{1}{\Gamma(r)} \lambda^r x^{r-1} e^{-\lambda x} & \text{for } x > 0, \\ 0 & \text{otherwise} \end{cases}$

$(X_1, \ldots, X_k) \sim \text{Multinomial}(n, k, p_1, \ldots, p_k)$,
joint PMF: $p(a_1, \ldots, a_k) = \binom{n}{a_1, \ldots, a_k} p_1^{a_1} \cdots p_k^{a_k}$ for $a_1, \ldots, a_k \ge 0$ with $a_1 + \cdots + a_k = n$.

Table of values for $\Phi(x)$, the CDF of a standard normal random variable

	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998

Second Midterm Exam

Name: _____

Please circle your section number!

Section 1	Section 2	Section 3	Section 4
Amy Cochran (MWF 9:55p)	Benedek Valkó (MWF 12:05p)	Sarah Tammen (TuTh 9:30a)	Shaoming Guo (MWF 8:50a)

- **There are 6 problems on the exam, some of them have multiple parts.** The problems are on pages 3 to 13. Make sure to read the problems carefully and budget your time wisely.
- If you are running out of time, please at least say *how* you would solve a given problem, preferably using notation, even if you do not have the time to do so. This may earn you partial credit.
- No calculators or other electronic devices, please. **Turn off your phone.** You do not have to carry out complicated numerical computations, but you should simplify your answer if it is possible with reasonable effort. In particular, geometric series must be evaluated.
- Your answer can be left in terms of the cumulative distribution function Φ of a standard normal random variable or in terms of the exponential function e^x, e.g. expressions like $\Phi(1)$ or $3e^{-9}$ would not need to be simplified further.
- Please present your solutions in a clear manner. **Justify your steps.** Numerical answers without explanation cannot get credit. Cross out the writing that you do not wish to be graded on.
- You can use the empty pages for additional space for writing if necessary.

Question	Points	Score
1	18	
2	20	
3	16	
4	12	
5	12	
6	22	
Total:	100	

List of named distributions

$X \sim \text{Bernoulli}(p)$, PMF: $p_X(0) = 1 - p, \quad p_X(1) = p$

$X \sim \text{Binomial}(n, p)$, PMF: $p_X(k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad 0 \le k \le n$

$X \sim \text{Geometric}(p)$, PMF: $p_X(k) = p(1-p)^{k-1}, \quad k = 1, 2, \ldots$

$X \sim \text{Poisson}(\lambda)$, PMF: $p_X(k) = \frac{\lambda^k}{k!} e^{-\lambda}, \quad k = 0, 1, 2, \ldots$

$X \sim \text{Hypergeometric}(N, N_A, n)$, PMF: $p_X(k) = \frac{\binom{N_A}{k}\binom{N-N_A}{n-k}}{\binom{N}{n}}, \quad 0 \le k \le n$

$X \sim \text{Uniform}(a, b)$, PDF: $f_X(x) = \begin{cases} \frac{1}{b-a} & \text{for } x \in [a, b], \\ 0 & \text{otherwise} \end{cases}$

$X \sim \text{Normal}(\mu, \sigma^2)$, PDF: $f_X(t) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(t-\mu)^2}{2\sigma^2}}, \quad t \in \mathbb{R}$

$X \sim \text{Exponential}(\lambda)$, PDF: $f_X(t) = \begin{cases} \lambda e^{-\lambda t} & \text{for } t > 0, \\ 0 & \text{otherwise} \end{cases}$

$(X_1, \ldots, X_k) \sim \text{Multinomial}(n, k, p_1, \ldots, p_k)$,
joint PMF: $p(a_1, \ldots, a_k) = \frac{n!}{a_1! a_2! \ldots a_k!} p_1^{a_1} \cdots p_k^{a_k}$ for $a_1, \ldots, a_k \ge 0$ with $a_1 + \cdots + a_k = n$.

1. **Short problems** The following short questions are not connected to each other. For each part, you have to *provide an appropriate explanation or justification.*

   (a) (6 points) The cumulative distribution of the random variable X is given by $F_X(x)$. Express the cumulative distribution function of $Y = 4X - 5$ in terms of F_X.

   (b) (6 points) Suppose that X has mean 1 and variance 2. Find numbers a, b so that $aX + b$ has mean zero and variance 8.

(Problem 1 continued)

(c) (6 points) The life length of a light bulb can be modeled with an exponential distribution, the expected life length is 500 days. What is the probability that the light bulb burns out before its expected life length?

2. The WW Candy Company is running a promotion for their candy bar. Each candy bar can contain a silver ticket (with probability 0.1), or a golden ticket (with probability 0.001), but not both.

   Note. You will approximate certain probabilities in some of the following problems. Justify the approach you use! You may assume that events related to different candy bars are independent.

   (a) (8 points) Augustus likes to eat candy, so he makes his parents buy 400 candy bars. Approximate the probability that he finds at least 31 silver tickets.

(Problem 2 continued)

(b) (8 points) Veruca would like to collect a golden ticket, so she makes her parents buy 500 candy bars. Approximate the probability that she finds at least one golden ticket among the purchased candy bars.

(c) (4 points) The WW Candy Company sells 10,000 candy bars in a week. Write down the *exact* probability that there are 2 golden tickets and 150 silver tickets among the sold candy bars.

3. We throw a dart uniformly at random at a disk shaped board with a radius of 10 inches. We then draw a circle that has the same center as the board and goes through the position of the dart. Let T be the area of the disk bounded by this random circle.

   (a) (2 points) What are the possible values of T?

   (b) (8 points) Find the cumulative distribution function of T.
   Note. Make sure that your function is defined for the whole real line!

(Problem 3 continued)

(c) (6 points) Explain why T is a continuous random variable, and find its probability density function.

4. The random variable X has moment generating function given by

$$M_X(t) = \frac{1}{4}e^{-t} + \frac{1}{2}e^{3t} + \frac{1}{8}e^{4t} + \frac{1}{8}e^{6t}.$$

(a) (6 points) Find the probability $P(X > 2)$.

(b) (6 points) Find the third derivative of M_X at $t = 0$ without differentiation.

5. Suppose that X and Y are independent random variables, with $X \sim \text{Geom}(1/2)$ and $Y \sim \text{Geom}(1/3)$.

(a) (4 points) Provide the joint probability mass function of (X, Y).

(b) (8 points) Find the probability $P(X = 2Y)$.

6. Suppose that (X, Y) are jointly continuous with joint probability density function

$$f(x,y) = \begin{cases} \frac{3}{\sqrt{2\pi}} e^{-\frac{x^2}{2} - 3(y-x)}, & \text{if } y \geq x \\ 0 & \text{otherwise.} \end{cases}$$

(a) (6 points) Find the marginal probability density function of X.

(b) (6 points) Decide whether X and Y are independent or not. Explain.
Hint. You can do this part without doing any of the other parts of this problem.

(Problem 6 continued)

(c) (4 points) Write an integral expression for the probability $P(X > 0, Y \leq 1)$.

Note. Do not attempt to compute the integral! Your answer should be the integral of an explicit (not case-defined!) function, make sure that the boundaries of integration are correct.

(Problem 6 continued)

(d) (6 points) Find the marginal probability density function of Y.
Hint. The answer will have the function Φ in it.

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Second Midterm Exam - Solutions

1. **Short problems** The following short questions are not connected to each other. For each part, you have to *provide an appropriate explanation or justification.*

   (a) (6 points) The cumulative distribution of the random variable X is given by $F_X(x)$. Express the cumulative distribution function of $Y = 4X - 5$ in terms of F_X.

   Solution: Using the definition of the CDF together with the definition of Y we get

$$F_Y(y) = P(Y \leq y) = P(4X - 5 \leq y) = P\left(X \leq \frac{y+5}{4}\right) = F_X\left(\frac{y+5}{4}\right)$$

   (b) (6 points) Suppose that X has mean 1 and variance 2. Find numbers a, b so that $aX + b$ has mean zero and variance 8.

   Solution: We have seen in class that for a, b real numbers and a random variable X we have

$$E[aX + b] = aE[X] + b, \qquad \mathrm{Var}(aX + b) = a^2 \,\mathrm{Var}(X).$$

   Hence if a, b satisfy the requirements of the problem then we have

$$0 = E[aX + b] = aE[X] + b = a + b$$

   and

$$8 = \mathrm{Var}(aX + b) = a^2 \,\mathrm{Var}(X) = 2a^2.$$

   The last equation gives $a = \pm 2$, from which we get the possible solutions $a = 2, b = -2$ and $a = -2, b = 2$. (Either one would be an acceptable solution.)

   (c) (6 points) The life length of a light bulb can be modeled with an exponential distribution, the expected life length is 500 days. What is the probability that the light bulb burns out before its expected life length?

   Solution: Denote the life length of the light bulb by X.

   The expected value of an $\mathrm{Exp}(\lambda)$ distributed random variable is $1/\lambda$ (we have seen this in class). Since $E[X] = 500$ and X is exponential, we get $X \sim \mathrm{Exp}(1/500)$. We need to compute $P(X < E[X]) = P(X < 500)$. Using the PDF of the exponential we get

$$P(X < 500) = \int_{-\infty}^{500} f(x)dx = \int_0^{500} \frac{1}{500} e^{x/500} dx = 1 - e^{-1}.$$

   The last step follows from computing the integral directly. We can also compute it by noting that $P(X < 500) = P(X \leq 500)$ (since X is continuous), and the CDF of the $\mathrm{Exp}(\lambda)$ distribution is $1 - e^{-\lambda x}$ for positive x, and 0 otherwise.

2. The WW Candy Company is running a promotion for their candy bar. Each candy bar can contain a silver ticket (with probability 0.1), or a golden ticket (with probability 0.001), but not both.

Note. You will approximate certain probabilities in some of the following problems. Justify the approach you use! You may assume that events related to different candy bars are independent.

(a) (8 points) Augustus likes to eat candy, so he makes his parents buy 400 candy bars. Approximate the probability that he finds at least 31 silver tickets.

Solution: Let X be the number of silver tickets that Augustus finds. Consider success to be a candy bar containing a silver ticket. Each candy bar has a silver ticket with probability .1. Since the events related to different candy bars are independent, we can view X as the number of successes within 400 independent trials with the same success probability, so it will have a binomial distribution with $n = 400$ and $p = .1$.

We want to approximate $P(X \geq 31)$. Checking that $np(1-p) = 400(.1)(0.9) = 36 > 10$, we justify the use of a normal random variable to approximate this probability. This approximation involves first standardizing X (i.e. subtract its mean $E[X] = np = 400(0.1) = 40$ and divide by its standard deviation $\sqrt{\text{Var}(X)} = \sqrt{np(1-p)} = \sqrt{36} = 6$):

$$P(X \geq 31) = P\left(\frac{X-40}{6} \geq \frac{31-40}{6}\right).$$

Letting $Z \sim \text{Normal}(0,1)$ with CDF Φ, then

$$P\left(\frac{X-40}{6} \geq \frac{31-40}{6}\right) \approx P\left(Z \geq \frac{31-40}{6}\right) = 1 - P\left(Z < -\frac{9}{6}\right) = 1 - \Phi(-3/2) = \Phi(3/2).$$

Above, we used the relation that $\Phi(z) = P(Z \leq z) = P(Z < z)$ (Z is continuous) and $\Phi(z) = 1 - \Phi(-z)$.

Alternatively, it was equally valid to approximate

$$P(X \geq 31) = 1 - P(X \leq 30) = 1 - P\left(\frac{X-40}{6} \leq \frac{30-40}{6}\right)$$

which leads to an answer of $1 - \Phi(-5/3) = \Phi(5/3)$.

(b) (8 points) Veruca would like to collect a golden ticket, so she makes her parents buy 500 candy bars. Approximate the probability that she finds at least one golden ticket among the purchased candy bars.

Solution: Let X be the number of golden tickets that Veruca finds. Following a similar argument as in (a), consider success to be a candy bar containing a golden ticket. Each candy bar has a golden ticket with probability .001. Since the events related to different candy bars are independent, we can view X as the number of successes within 500 independent trials with the same success probability, so it will have a binomial distribution with $n = 500$ and $p = .001$.

We want to approximate $P(X \geq 1)$. Checking that $np^2 = .0005$ is small (i.e. much smaller than 1 or $<< 1$), we justify the use of a Poisson random variable with the same mean as X, which is $np = 500(.001) = .5$, to approximate this probability. Letting $Y \sim \text{Poisson}(.5)$, we therefore have

$$P(X \geq 1) \approx P(Y \geq 1) = 1 - P(Y = 0) = 1 - e^{-.5}.$$

For the last equality, we use the PMF of a Poisson random variable with parameter .5.

(c) (4 points) The WW Candy Company sells 10,000 candy bars in a week. Write down the *exact* probability that there are 2 golden tickets and 150 silver tickets among the sold candy bars.

Solution: Let X be the number of golden tickets, Y be the number of silver tickets, and Z be the number of bars without golden or silver tickets among the 10,000 candy bars. No candy bar yields both a golden and a silver ticket, which means that $X + Y + Z = 10000$. Since the events related to each bar are independent, we view each bar as an independent trial with three possible outcomes (golden ticket, silver ticket, or neither). The respective probabilities of these outcomes are .001, .1, and $1 - .001 - 0.1 = .899$. Thus, the random vector (X, Y, Z) is Multinomial(n, r, p_1, p_2, p_3) with $n = 10000$, $r = 3$, $p_1 = .001$, $p_2 = .1$, $p_3 = .899$. Using the joint PMF of a multinomial random variable,

$$P(X = 2, Y = 150) = P(X = 2, Y = 150, Z = 9848) = \frac{10000!}{2!150!9848!}.001^2.1^{150}.899^{9848}.$$

Above, we use the fact that the event that there are 2 golden tickets and 150 tickets is exactly the event that there are 2 golden tickets, 150 silver tickets, and 9848 bars without any tickets.

3. We throw a dart uniformly at random at a disk shaped board with a radius of 10 inches. We then draw a circle that has the same center as the board and goes through the position of the dart. Let T be the area of the disk bounded by this random circle.

(a) (2 points) What are the possible values of T?

Solution: Let R denote the distance of the dart from the origin, then we have $0 \leq R \leq 10$. The value of T is the area of the disk with radius R: $T = \pi R^2$. Since $0 \leq R \leq 10$, we have $0 \leq T \leq 100\pi$.

(b) (8 points) Find the cumulative distribution function of T.
Note. Make sure that your function is defined for the whole real line!

Solution: We need to compute $F_T(t) = P(T \leq t)$ for all values of t. Since T can only take values from $[0, 100\pi]$, we have

$$F_T(t) = 0, \qquad \text{for } t < 0$$

and

$$F_T(t) = 1, \qquad \text{for } t \geq 100\pi.$$

Now suppose that $t \in [0, 100\pi]$. The event $T \leq t$ in this case is the same as the dart landing inside the disk with area t. The probability of this is exactly $\frac{t}{100\pi}$ (since the position of the dart is a uniformly chosen point in the disk with area 100π). This means that the CDF is given by the following case-defined function:

$$F_T(t) = \begin{cases} 0, & \text{if } t < 0 \\ \frac{t}{100\pi}, & \text{if } 0 \leq t < 100\pi \\ 1, & \text{if } t \geq 100\pi \end{cases}$$

(c) (6 points) Explain why T is a continuous random variable, and find its probability density function.

Solution: The CDF found in part (a) is continuous everywhere: this is true because it is continuous within the three cases, and it is continuous at the boundaries $t = 0$ and $t = 100\pi$. The CDF is differentiable everywhere outside of $t = 0$ and $t = 100\pi$, from this it follows that T has continuous distribution. We can obtain the PDF of T by differentiating the CDF (where it is differentiable), which gives

$$f_T(t) = \begin{cases} \frac{1}{100\pi}, & 0 < t < 100\pi \\ 0, & \text{otherwise.} \end{cases}$$

Note that this is just the Unif$(0, 100\pi)$ distribution.

4. The random variable X has moment generating function given by

$$M_X(t) = \frac{1}{4}e^{-t} + \frac{1}{2}e^{3t} + \frac{1}{8}e^{4t} + \frac{1}{8}e^{6t}.$$

(a) (6 points) Find the probability $P(X > 2)$.

Solution: From the form of the MGF, we can recognize that it can be obtained from a discrete distribution: the exponents identify the possible values $-1, 3, 4, 6$, the coefficients identify the corresponding probabilities:

$$P(X = -1) = \frac{1}{4}, \quad P(X = 3) = \frac{1}{2}, \quad P(X = 4) = \frac{1}{8}, \quad P(X = 6) = \frac{1}{8}.$$

This discrete random variable has the appropriate MGF (from the formula $M_X(t) = \sum_k e^{kt}P(X = k)$), and since the MGF is finite in the neighborhood of 0 (in fact, everywhere), this means that this is the only distribution with this MGF.

We can compute $P(X > 2)$ by collecting the possible values of X larger than 2, and adding up the corresponding probabilities:

$$P(X > 2) = P(X = 3) + P(X = 4) + P(X = 6) = \frac{1}{2} + \frac{1}{8} + \frac{1}{8} = \frac{3}{4}.$$

(b) (6 points) Find the third derivative of M_X at $t = 0$ without differentiation.

Solution: Since the MGF is finite in a neighborhood of 0, we can compute the moments of X from the appropriate derivative of M_X at 0. In particular,

$$M_X'''(0) = E[X^3],$$

which we can compute from the PMF found in part (a):

$$E[X^3] = \sum_k k^3 P(X = k) = (-1)^3\frac{1}{4} + 3^3\frac{1}{2} + 4^3\frac{1}{8} + 6^3\frac{1}{8}.$$

5. Suppose that X and Y are independent random variables, with $X \sim$ Geom$(1/2)$ and $Y \sim$ Geom$(1/3)$.

(a) (4 points) Provide the joint probability mass function of (X, Y).

Solution: Because of independence we get

$$P(X = k, Y = \ell) = P(X = k)P(Y = \ell) = (1/2)(1/2)^{k-1}(1/3)(2/3)^{\ell-1}$$

for $k, \ell = 1, 2, \ldots$. Here we used the PMF of Geom(p) for $p = 1/2$ and $p = 1/3$.

(b) (8 points) Find the probability that $P(X = 2Y)$.

Solution: We first break up the event $\{X = 2Y\}$ into smaller disjoint events based on the possible values of Y:

$$P(X = 2Y) = \sum_{k=1}^{\infty} P(X = 2k, Y = k).$$

Using part (a), we get

$$P(X = 2Y) = \sum_{k=1}^{\infty} P(X = 2k)P(Y = k) = \sum_{k=1}^{\infty}(1/2)^{2k}(1/3)(2/3)^{k-1}.$$

The last sum is just an infinite geometric series:

$$\sum_{k=1}^{\infty}(1/2)^{2k}(1/3)(2/3)^{k-1} = \frac{1}{4\cdot 3}\sum_{k=1}^{\infty}\left(\tfrac{1}{4}\cdot\tfrac{2}{3}\right)^{k-1} = \frac{1}{12}\sum_{k=1}^{\infty}\left(\tfrac{1}{6}\right)^{k-1} = \frac{1}{12}\cdot\frac{1}{1-1/6} = \frac{1}{10}.$$

Hence $P(X = 2Y) = \frac{1}{10}$.

6. Suppose that (X, Y) are jointly continuous with joint probability density function

$$f(x,y) = \begin{cases} \frac{3}{\sqrt{2\pi}}e^{-x^2/2-3(y-x)}, & \text{if } y \geq x \\ 0 & \text{otherwise.} \end{cases}$$

(a) (6 points) Find the marginal probability density function of X.

Solution: We can find the marginal PDF of X by integrating out the y variable from the joint PDF:

$$f_X(x) = \int_{-\infty}^{\infty} f(x,y)dy$$

For a fixed x the value of $f(x,y)$ is positive for $y \geq x$, hence

$$\begin{aligned} f_X(x) &= \int_{-\infty}^{\infty} f(x,y)dy = \int_{x}^{\infty} \frac{3}{\sqrt{2\pi}}e^{-x^2/2-3(y-x)}dy = \frac{3}{\sqrt{2\pi}}e^{-x^2/2+3x}\int_{x}^{\infty} e^{-3y}dy \\ &= \frac{3}{\sqrt{2\pi}}e^{-x^2/2+3x}\frac{1}{3}e^{-3x} = \frac{1}{\sqrt{2\pi}}e^{-x^2/2}. \end{aligned}$$

The integral can be computed directly, or by noting that $\int_x^{\infty} 3e^{-3(y-x)}dy = \int_0^{\infty} 3e^{-3t}dt = 1$ (after a change of variables and noting that we are integrating a PDF).

Note that X is a standard normal random variable.

(b) (6 points) Decide whether X and Y are independent or not. Explain.

Hint. You can do this part without doing any of the other parts of this problem.

Solution: If X and Y are jointly continuous, then they are independent exactly if the joint PDF is the product of the marginal PDFs: $f(x,y) = f_X(x)f_Y(y)$. From the form of the joint PDF, we see that the marginals are both positive: if we fix x or y and integrate in the other variable then we integrate a function that is positive on a semi-infinite interval: (x, ∞) and $(-\infty, y)$, respectively. Hence $f_X(x)f_Y(y)$ is positive for all x, y, which means that it cannot be equal to $f(x,y)$ everywhere, as $f(x,y) = 0$ for $y < x$. Hence X and Y are not independent.

Note: there are lots of other ways that one can show this statement. E.g. using the result of (a) we see that $\frac{f(x,y)}{f_X(x)}$ is *not* a function that only depends on y, which also shows that we cannot $f(x,y) = f_X(x)f_Y(y)$ everywhere.

(c) (4 points) Write an integral expression for the probability $P(X > 0, Y \leq 1)$.

Note. Do not attempt to compute the integral! Your answer should be the integral of an explicit (not case-defined!) function, make sure that the boundaries of integration are correct.

Solution: Since X, Y are jointly continuous, we have

$$P(X > 0, Y \leq 1) = \iint_{x>0, y\leq 1} f(x,y)dxdy.$$

We need to identify the region where $x > 0, y \leq 1$ and $y \geq x$ (since $f(x,y) = 0$ if $y < x$). From the last two inequalities, we get $x \leq y \leq 1$ which also gives $x \leq 1$. Together with the first inequality, we get $0 < x \leq 1$. Hence, for a fixed $0 < x \leq 1$ the integration boundaries for y are x and 1:

$$P(X > 0, Y \leq 1) = \int_0^1 \int_x^1 \frac{3}{\sqrt{2\pi}} e^{-x^2/2-3(y-x)} dy dx.$$

If we fix y first, then $0 \leq y \leq 1$ and $0 \leq x \leq y$:

$$P(X > 0, Y \leq 1) = \int_0^1 \int_0^y \frac{3}{\sqrt{2\pi}} e^{-x^2/2-3(y-x)} dx dy.$$

(d) (6 points) Find the marginal probability density function of Y.
Hint. The answer will have the function Φ in it.

Solution: We get the marginal of Y by integrating out x in the joint density:

$$f_Y(y) = \int_{-\infty}^{\infty} f(x,y) dx = \int_{-\infty}^{y} \frac{3}{\sqrt{2\pi}} e^{-x^2/2-3(y-x)} dx$$

Moving the y dependent terms outside, and completing the square in the exponent gives

$$f_Y(y) = 3e^{-3y} \int_{-\infty}^{y} \frac{1}{\sqrt{2\pi}} e^{-x^2/2+3x} dx = 3e^{-3y} \int_{-\infty}^{y} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(x-3)^2+\frac{9}{2}} dx = 3e^{-3y+\frac{9}{2}} \int_{-\infty}^{y} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(x-3)^2} dx$$

In the integral, we can recognize the PDF of a $N(3,1)$ random variable, hence

$$\int_{-\infty}^{y} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(x-3)^2} dx = P(Y \leq y)$$

with $Y \sim N(3,1)$. But then $Z = Y - 3 \sim N(0,1)$:

$$\int_{-\infty}^{y} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(x-3)^2} dx = P(Y \leq y) = P(Y - 3 \leq y - 3) = P(Z \leq y - 3) = \Phi(y-3).$$

Collecting the terms gives

$$f_Y(y) = 3e^{-3y+\frac{9}{2}} \Phi(y-3).$$

Department of Mathematics, University of Wisconsin-Madison

Math 431 — Midterm Exam 2 — Solutions — Fall 2024

NAME : (as it appears on Canvas)

EMAIL: @wisc.edu

PROFESSOR: David Clancy or Mikhail Ivanov

Time: **90 minutes**

- This exam contains 5 questions some with multiple parts, 11 pages (including the cover) for the total of 93 points. Read the problems carefully and budget your time wisely.
- You are allowed a single sheet of hand-written notes.
- **NO CALCULATORS** or other electronic devices are to be used. Turn off your phone so as to not disturb others.
- You **do not** need to simplify binomial coefficients or factorials.
- Please present your solutions in a clear manner. Cross out any writing that you do not wish to be graded.
- Justify your steps.
- For binomial random variable approximations, you will NOT get full credit for your answer if you do not include some explanation of why that approximation is valid.
- You can use the attached table of the values of the $\Phi(x)$ function.
- If you use an additional page for a particular problem, be sure to **CLEARLY** indicate this on the problem's page so I know to look further.

Question:	1	2	3	4	5	Total
Points:	16	29	8	30	10	93
Score:						

1. The following questions are **Answers Only**.

(a) (4 points) A random variable Y has moment generating function

$$M_Y(t) = \frac{1}{15} + \frac{1}{3}e^{2t} + \frac{1}{5}e^{3t} + \frac{1}{3}e^{4t} + \frac{1}{15}e^{7t}.$$

What is the probability that Y is even?

Solution:

$$P(Y \text{ is even}) = P(Y=0) + P(Y=2) + P(Y=4) = \frac{1}{15} + \frac{1}{3} + \frac{1}{3} = \frac{11}{15}.$$

(b) (4 points) Let $X \sim \text{Exp}(1/3)$. What is $P(X < 5 | X \geq 3)$?

Solution:

$$P(X < 5|X \geq 3) = 1 - P(X \geq 3 + 2|X \geq 3) = 1 - P(X \geq 2) = 1 - e^{-\frac{2}{3}}.$$

Or

$$P(X < 5|X \geq 3) = \frac{P(3 \leq X < 5)}{P(X \geq 3)} = \frac{e^{-1} - e^{-5/3}}{e^{-1}} = 1 - e^{-2/3}.$$

(c) (4 points) The joint probability mass function of the random variables (X, Y) is given below:

		Y			
		0	1	2	3
X	0	1/10	1/5	3/10	0
	1	1/5	0	1/10	1/10

What is the marginal probability mass function of Y?

Solution:

Y			
0	1	2	3
3/10	1/5	2/5	1/10

(d) (4 points) Find $\text{Var}(X)$ where $X \sim \text{Unif}\{1, 2, 3, 4\}$.

Solution:

$$\text{Var}(X) = E(X^2) - E(X)^2 = \frac{1}{4}(1+4+9+16) - \left(\frac{1+2+3+4}{4}\right)^2 = 7.5 - 6.25 = 1.25.$$

2. Let X and Y have joint pdf

$$f_{X,Y}(x,y) = \begin{cases} x+y, & \text{for } 0<x<1 \text{ and } 0<y<1 \\ 0 & \text{otherwise.} \end{cases}$$

(a) (5 points) Check that this is a valid joint pdf.

Solution: $f(x,y) = f_{X,Y}(x,y) \geq 0$ for all x, y and

$$\begin{aligned} \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\,dx\,dy &= \int_{y=0}^{1}\int_{x=0}^{1} (x+y)\,dx\,dy \\ &= \int_{y=0}^{1} \left(\frac{1}{2}x^2 + xy\right)\Big|_{x=0}^{x=1} dy = \int_0^1 \frac{1}{2} + y\,dy \\ &= \frac{1}{2}(y+y^2)\Big|_0^1 = \frac{1}{2} + \frac{1}{2} = 1 \end{aligned}$$

(b) (5 points) Find the marginal pdfs of X and Y.

Solution: The marginals

$$f_X(x) = \int_{y=0}^{1} (x+y)\,dy = xy + \frac{1}{2}y^2\Big|_{y=0}^{y=1} = x + \frac{1}{2} \text{ for } x \in (0,1)$$

otherwise $f_X(x) = 0$. We computed in part (a)

$$f_Y(y) = \int_{x=0}^{1} (x+y)\,dx = \frac{1}{2} + y \qquad \text{for } y \in (0,1).$$

Thus

$$f_X(x) = \begin{cases} \frac{1}{2} + x & : 0<x<1 \\ 0 & : \text{else} \end{cases} \qquad f_Y(y) = \begin{cases} \frac{1}{2} + y & : 0<y<1 \\ 0 & : \text{else} \end{cases}.$$

(c) (5 points) Are X and Y independent?

Solution: No. For $x, y \in (0,1)$ we have

$$f_X(x)f_Y(y) = \frac{1}{4} + \frac{1}{2}(x+y) + xy \neq x + y = f_{X,Y}(x,y).$$

(problem 2 continues)
Let X and Y have joint pdf

$$f_{X,Y}(x,y) = \begin{cases} x+y, & \text{for } 0 < x < 1 \text{ and } 0 < y < 1 \\ 0 & \text{otherwise.} \end{cases}$$

(d) (7 points) Find $P(X < Y)$.

Solution:

$$P(X < Y) = \iint_{x<y} f_{X,Y}(x,y)\,dx\,dy = \int_{y=0}^{1}\int_{x=0}^{y}(x+y)\,dx\,dy$$
$$= \int_{y=0}^{1} \frac{1}{2}x^2 + xy\Big|_{x=0}^{x=y} dy = \int_0^1 \frac{3}{2}y^2\,dy = \frac{1}{2}y^3\Big|_{y=0}^{1} = \frac{1}{2}.$$

(e) (7 points) Find $E[XY]$.

Solution:

$$E[XY] = \int_{y=0}^{1}\int_{x=0}^{1}(xy)(x+y)\,dx\,dy = \int_{y=0}^{1}\int_{x=0}^{1}(x^2y + xy^2)\,dx\,dy = \int_0^1 \frac{1}{3}x^3y + \frac{1}{2}x^2y^2\Big|_{x=0}^{x=1} dy$$
$$= \int_0^1 \frac{1}{3}y + \frac{1}{2}y^2\,dy = \frac{1}{6}y^2 + \frac{1}{6}y^3\Big|_{y=0}^{1} = \frac{1}{3}.$$

3. (8 points) Let X be a random variable with probability density function

$$f_X(x) = \begin{cases} 8x^{-3} & \text{for } x \geq 2 \\ 0 & \text{else.} \end{cases}$$

Let $Y = X^3$. What is the probability density function of Y?

Solution: The CDF of Y is

$$P(Y \leq t) = P(X^3 \leq t) = P(X \leq t^{1/3}).$$

For $t^{1/3} < 2$, i.e. $t < 8$ $P(X \leq t) = 0$. For $t \geq 8$

$$P(Y \leq t) = \int_2^{t^{1/3}} 8x^{-3}\,dx = -4x^{-2}\Big|_{x=2}^{x=t^{1/3}} = 1 - 4t^{-2/3}.$$

So

$$\frac{d}{dt}P(Y \leq t) = \begin{cases} 0 & : t < 8 \\ \text{undefined} & : t = 8 \\ \frac{8}{3}t^{-5/3} & : t > 8. \end{cases} \qquad \text{and} \qquad f_Y(t) = \begin{cases} 0 & : t < 8 \\ \frac{8}{3}t^{-5/3} & : t \geq 8. \end{cases}$$

4. Each day, John plays 7 games of chess. He wins, draws, or loses each game independently with probabilities 1/2, 1/6, 1/3.

   (a) (6 points) Compute the exact probability that John wins 7 times, draws 5 times and loses 2 times this weekend.

Solution: This is a multinomial $\text{Mult}(14, 3, (1/2, 1/6, 1/3))$ so

$$P(7 \text{ wins}, 5 \text{ draws}, \ 2 \text{ losses}) = \binom{14}{7,5,2} \left(\frac{1}{2}\right)^7 \left(\frac{1}{6}\right)^5 \left(\frac{1}{3}\right)^2$$

$$= \frac{14!}{7!\,5!\,2!} \left(\frac{1}{2}\right)^7 \left(\frac{1}{6}\right)^5 \left(\frac{1}{3}\right)^2.$$

   (b) (8 points) John calls a day "great" if he wins all 7 of the games he plays that day. Approximate that probability that over the next $256 (= 2^8)$ days John has 3 great days.

Solution: The probability of a great day is $(1/2)^7 = 2^{-7}$. So

$$P(3 \text{ great days over } 256 \text{ days}) = P(\text{Bin}(2^8, 2^{-7}) = 3).$$

We can use Poisson approximation because $np^2 = 2^8 \times 2^{-14} = 2^{-6} = \frac{1}{64} \leq \frac{1}{10}$. So

$$P(\text{Bin}(2^8, 2^{-7}) = 3) \approx P(\text{Poisson}(2 = np) = 3) = \frac{e^{-2}2^3}{3!}.$$

(c) (8 points) John calls a day "good" if he wins at least half of his games (i.e. he wins 4, 5, 6, 7). The probability that a day is good is $\frac{1}{2}$ (you do **NOT** need to check it). Approximate the probability that over the next 256 days the number of good days is between 116 and 132 (inclusive).

Solution: Let S_n be the number of good days over the next 256 days. Then $S_n \sim \text{Bin}(2^8, 1/2)$. So $np(1-p) = 2^6 = 64 > 10$ and therefore the normal approximation to the binomial is good.
We compute $np = 2^7 = 128$ and $np(1-p) = 2^6$ so $\sqrt{np(1-p)} = 2^3 = 8$.

$$\begin{aligned} P(116 \le S_n \le 132) &= P\left(\frac{116-128}{8} \le \frac{S_n - np}{\sqrt{np(1-p)}} \le \frac{132-128}{8}\right) \\ &\approx P\left(\frac{-12}{8} \le Z \le \frac{4}{8}\right) = P(-1.5 \le Z \le .5) = \Phi(.5) - \Phi(-1.5) \\ &= \Phi(.5) - (1 - \Phi(1.5)) \approx 0.6915 + 0.9332 - 1 = 0.6247. \end{aligned}$$

With continuity correction you get

$$P \approx 0.6529.$$

(d) (8 points) John's friend Mark is skeptical that John actually only loses 1/3 of his games. Mark notices that over the last 4 weeks (=28 days) John lost 80 of the $196 (= 2^2 \times 7^2)$ games he played. Does $p = 1/3 \approx 0.333$ land in the 95% interval of $\widehat{p} = 80/196 \approx 0.408$?

Solution: The 95% confidence interval is $(\widehat{p} - \varepsilon, \widehat{p} + \varepsilon)$

$$\begin{aligned} 2\Phi(2\varepsilon\sqrt{n}) - 1 &= 0.95 \\ \Phi(2\varepsilon\sqrt{196}) &= 0.975 \\ 28\varepsilon &= 1.96 \\ \varepsilon &= \frac{1.96}{2^2 \times 7} = \frac{196/100}{2^2 \times 7} = \frac{7}{100} = 0.07. \end{aligned}$$

Hence

$$(\widehat{p} - \varepsilon, \widehat{p} + \varepsilon) = (0.338, 0.478).$$

So $\widehat{p} - \varepsilon \approx 0.338 > 1/3$. So p is **not** in the confidence interval.

5. Suppose that X is a random variable with moment generating function

$$M_X(t) = \begin{cases} \dfrac{25}{(5-t)^2} & : t < 5 \\ \infty & : \text{else} \end{cases}.$$

(a) (5 points) Compute the mean of X.

Solution: Write $M = M_X$. The mean is

$$E[X] = M'(0) = \left.\frac{50}{(5-t)^3}\right|_{t=0} = \frac{50}{5^3} = \frac{2}{5}.$$

(b) (5 points) Compute the variance of X.

Solution: The second moment is

$$E[X^2] = M''(0) = \left.\frac{150}{(5-t)^4}\right|_{t=0} = \frac{150}{5^4} = \frac{6}{25}.$$

So the variance is

$$\mathrm{Var}(X) = E[X^2] - E[X]^2 = \frac{6}{25} - \frac{4}{25} = \frac{2}{25}.$$

Solution:

$$M_X(t) = \frac{5}{5-t} \cdot \frac{5}{5-t},$$

and we can recognize $\frac{5}{5-t}$ near $t = 0$ as a moment generation function of Exp(5) distribution. So, given independent $X_1, X_2 \sim \mathrm{Exp}(\lambda)$, we have $X \sim X_1 + X_2$, so

$$E(X) = E(X_1 + X_2) = E(X_1) + E(X_2) = \frac{1}{5} + \frac{1}{5} = \frac{2}{5}$$
$$\mathrm{Var}(X) = \mathrm{Var}(X_1 + X_2) = \mathrm{Var}(X_1) + \mathrm{Var}(X_2) = \frac{1}{25} + \frac{1}{25} = \frac{2}{25}$$

SCRATCH PAPER - DO NOT REMOVE FROM YOUR EXAM.
SCRATCH WORK WILL NOT BE GRADED

SCRATCH PAPER - DO NOT REMOVE FROM YOUR EXAM
SCRATCH WORK WILL NOT BE GRADED

First Name: _____ Last Name: _____

Table of values for $\Phi(x)$, the CDF of a standard normal random variable

	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998

Department of Mathematics, University of Wisconsin-Madison

Math 431 — Midterm Exam 2 — Spring 2025

NAME : (as it appears on Canvas)

EMAIL: @wisc.edu

PROFESSOR: Mikhail Feldman or Mikhail Ivanov

INSTRUCTIONS:

Time: **90 minutes**

- This exam contains 7 questions some with multiple parts, 12 pages (including the cover) for the total of 83 points. Read the problems carefully and budget your time wisely.
- You are allowed a single sheet of hand-written notes.
- **NO CALCULATORS** or other electronic devices are to be used. Turn off your phone so as to not disturb others.
- You **do not** need to simplify binomial coefficients or factorials, except when you are asked to do it.
- Please present your solutions in a clear manner. Cross out any writing that you do not wish to be graded. Justify your steps.
- Remember, expectation and variance can be either finite, infinite or undefined.
- If you use an additional page for a particular problem, be sure to **CLEARLY** indicate this on the problem's page so I know to look further.

Question:	1	2	3	4	5	6	7	Total
Points:	16	15	8	6	12	18	8	83
Score:								

1. (16 points) **Short problems. (4 points each)** In the following problems (but only here!) you will not need to justify your answer.

   (a) Suppose that a random variable X has mean 2 and variance 3. Find numbers a, b so that $aX + b$ has mean 1 and variance 12.

   (b) In humans (and many other organisms), genes come in pairs. Consider a gene of interest, which comes in two types (alleles): type a and type A. The genotype of a person for that gene is the types of the two genes in the pair: AA, Aa, or aa (aA is equivalent to Aa). According to the Hardy–Weinberg law, for a population in equilibrium, the frequencies of AA, Aa, aa will be p^2, $2p(1-p)$, $(1-p)^2$, respectively, for some p with $0 < p < 1$. Suppose that the Hardy–Weinberg law holds, and that n people are drawn randomly from the population, independently. Let X_1, X_2, X_3 be the number of people in the sample with genotypes AA, Aa, aa, respectively. What is the joint PMF of X_1, X_2, X_3?

(c) Suppose the duration of a phone call is given by an exponential distribution with an expected length of 15 minutes. Determine the conditional probability that a call lasts more than 20 minutes, given that it has already lasted for 10 minutes.(You don't have to simplify the answer.)

(d) Let X be a random variable with moment generating function

$$M(t) = \begin{cases} \dfrac{1}{\sqrt{1-2t}}, & |t| < \frac{1}{2}. \\ \infty, & \text{otherwise.} \end{cases}$$

Find the expectation $E[X]$, and the variance $\mathrm{Var}[X]$.

2. Let X and Y be discrete random variables defined on the same sample space. The table below gives the values of the joint probability mass function of X, Y, for example, $p_{X,Y}(1,-1) = P(X=1, Y=-1) = 1/8$.

		Y		
		-1	0	1
X	1	$\frac{1}{8}$	0	$\frac{2}{8}$
	2	$\frac{1}{8}$	$\frac{2}{8}$	$\frac{2}{8}$

(a) (4 points) Find the marginal probability mass functions of X and Y.

(b) (4 points) Are X and Y independent? Give a rigorous explanation.

(c) (4 points) Find $\mathrm{Var}(Y)$.

(d) (3 points) Find $P(Y < X)$.

3. (8 points) Let $X \sim \text{Unif}\,[0, 5]$. Find the probability density function of the random variable $Y = X^2$.

4. (6 points) The owner of a certain website is studying the distribution of the number of visitors to the site. Every day, a million $(1,000,000)$ people independently decide whether to visit the site, with probability $p = 2 \cdot 10^{-6}$ of visiting. Give a good approximation for the probability of getting at least three visitors on a particular day.

5. Let random variables X and Y have joint pdf

$$\begin{cases} A(x,y) & \text{for } 0 \le x \le 4, 0 \le y \le 3, \\ 0 & \text{otherwise} \end{cases}$$

Your integrals in the answers in this problem should be of the form $\int_a^b \int_c^d \ldots$ with appropriate limits of integration a, b, c, d and an explicit function inside the integral.

(a) (4 points) Which conditions do we need to check to be sure that this is actual joint pdf?

(b) (4 points) Write an integral expression for the $E[X^2Y]$.

(c) (4 points) Write an integral expression for the $P(X + Y < 2)$.

6. Suppose random variable X has density function

$$f_X(x) = \begin{cases} \dfrac{2}{x^3}, & x \geq 1 \\ 0, & x < 1 \end{cases}$$

(a) (6 points) Find $P(2 \leq X \leq 3)$

(b) (6 points) Find $E[X]$

(c) (6 points) Find $\mathrm{Var}[X]$

7. (8 points) Our ice machines produce ice cubes, 3 or 4 ice cubes every second with equal probabilities. Approximate the probability that that in 1 minutes and 40 seconds we will have at least 360 ice cubes.

First Name: _____ Last Name: _____

Table of values for $\Phi(x)$, the CDF of a standard normal random variable

	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998

SCRATCH PAPER - DO NOT REMOVE FROM YOUR EXAM.
SCRATCH WORK WILL NOT BE GRADED

Department of Mathematics, University of Wisconsin-Madison

Math 431 — Midterm Exam 2 — Solutions — Spring 2025

NAME : (as it appears on Canvas)

EMAIL: @wisc.edu

PROFESSOR: Mikhail Feldman or Mikhail Ivanov

INSTRUCTIONS:

Time: **90 minutes**

- This exam contains 7 questions some with multiple parts, 12 pages (including the cover) for the total of 83 points. Read the problems carefully and budget your time wisely.
- You are allowed a single sheet of hand-written notes.
- **NO CALCULATORS** or other electronic devices are to be used. Turn off your phone so as to not disturb others.
- You **do not** need to simplify binomial coefficients or factorials, except when you are asked to do it.
- Please present your solutions in a clear manner. Cross out any writing that you do not wish to be graded. Justify your steps.
- Remember, expectation and variance can be either finite, infinite or undefined.
- If you use an additional page for a particular problem, be sure to **CLEARLY** indicate this on the problem's page so I know to look further.

Question:	1	2	3	4	5	6	7	Total
Points:	16	15	8	6	12	18	8	83
Score:								

1. (16 points) **Short problems. (4 points each)** In the following problems (but only here!) you will not need to justify your answer.

   (a) Suppose that a random variable X has mean 2 and variance 3. Find numbers a, b so that $aX + b$ has mean 1 and variance 12.

Solution: We have seen in class that for a, b real numbers and a random variable X we have

$$E[aX + b] = aE[X] + b, \qquad \mathsf{Var}(aX + b) = a^2\mathsf{Var}(X).$$

Hence if a, b satisfy the requirements of the problem, then we have

$$1 = E[aX + b] = aE[X] + b = 2a + b$$

and

$$12 = \mathsf{Var}(aX + b) = a^2\mathsf{Var}(X) = 3a^2.$$

The last equation gives $a = \pm 2$, from which, using the first equation $2a + b = 1$, we get the possible solutions $a = 2, b = -3$ and $a = -2, b = 5$. (Either one would be an acceptable solution.)

   (b) In humans (and many other organisms), genes come in pairs. Consider a gene of interest, which comes in two types (alleles): type a and type A. The genotype of a person for that gene is the types of the two genes in the pair: AA, Aa, or aa (aA is equivalent to Aa). According to the Hardy–Weinberg law, for a population in equilibrium, the frequencies of AA, Aa, aa will be p^2, $2p(1-p)$, $(1-p)^2$, respectively, for some p with $0 < p < 1$. Suppose that the Hardy–Weinberg law holds, and that n people are drawn randomly from the population, independently. Let X_1, X_2, X_3 be the number of people in the sample with genotypes AA, Aa, aa, respectively. What is the joint PMF of X_1, X_2, X_3?

Solution:

$$(X_1, X_2, X_3) \sim Multinomial(n, 3, p^2, 2p(1-p), (1-p)^2),$$

so

$$p_{X_1,X_2,X_3}(k_1, k_2, k_3) = \frac{n!}{k_1!k_2!k_3!} \left(p^2\right)^{k_1} \left(2p(1-p)\right)^{k_2} \left((1-p)^2\right)^{k_3}$$

when k_i are non-negative integers satisfying $k_1 + k_2 + k_3 = n$, and 0 otherwise.

(c) Suppose the duration of a phone call is given by an exponential distribution with an expected length of 15 minutes. Determine the conditional probability that a call lasts more than 20 minutes, given that it has already lasted for 10 minutes.(You don't have to simplify the answer.)

Solution: Denote by T the length of phone call. The expected value of an $\text{Exp}(\lambda)$ is $1/\lambda$, hence $\lambda = 1/15$. By the memoryless property of the exponential distribution

$$P(T > 20|T > 10) = P(T > 10) = 1 - (1 - e^{-\lambda \cdot 10}) = e^{-\frac{10}{15}} = e^{-2/3}.$$

(d) Let X be a random variable with moment generating function

$$M(t) = \begin{cases} \dfrac{1}{\sqrt{1-2t}}, & |t| < \frac{1}{2}. \\ \infty, & \text{otherwise.} \end{cases}$$

Find the expectation $E[X]$, and the variance $\text{Var}[X]$.

Solution:

$$E[X] = M'(0) = -\frac{1}{2}(1-2t)^{-3/2}(-2)\Big|_{t=0} = 1.$$

$$E[X^2] = M''(0) = (M')'(0) = ((1-2t)^{-3/2})'\Big|_{t=0} = -\frac{3}{2}(1-2t)^{-5/2}(-2)\Big|_{t=0} = 3.$$

$$\text{Var}[X] = E[X^2] - (E[X])^2 = 3 - 1^2 = 2.$$

2. Let X and Y be discrete random variables defined on the same sample space. The table below gives the values of the joint probability mass function of X, Y, for example, $p_{X,Y}(1,-1) = P(X=1, Y=-1) = 1/8$.

		Y		
		-1	0	1
X	1	$\frac{1}{8}$	0	$\frac{2}{8}$
	2	$\frac{1}{8}$	$\frac{2}{8}$	$\frac{2}{8}$

(a) (4 points) Find the marginal probability mass functions of X and Y.

Solution: To find the marginal probabilities of X and Y, we need to sum the values of the joint pmf over rows and columns:

$$P(X=1) = \frac{1}{8} + 0 + \frac{2}{8} = \frac{3}{8}$$
$$P(X=2) = \frac{1}{8} + \frac{2}{8} + \frac{2}{8} = \frac{5}{8}$$

and

$$P(Y=-1) = \frac{1}{8} + \frac{1}{8} = \frac{2}{8}$$
$$P(Y=0) = 0 + \frac{2}{8} = \frac{2}{8}$$
$$P(Y=1) = \frac{2}{8} + \frac{2}{8} = \frac{4}{8}$$

(b) (4 points) Are X and Y independent? Give a rigorous explanation.

Solution: X and Y are not independent. Indeed, if they were independent we would have $P(X=k_1, Y=k_2) = P(X=k_1)P(Y=k_2)$ for all $k_1 \in \{1,2\}$ and all $k_2 \in \{-1,0,1\}$, but we have

$$P(X=1, Y=0) = 0 \qquad \text{and} \qquad P(X=1)P(Y=0) = \frac{3}{8} \cdot \frac{2}{8} > 0.$$

Hence, $P(X=1, Y=0) \neq P(X=1)P(Y=0)$.

(c) (4 points) Find $\mathrm{Var}(Y)$.

Solution: We have

$$E[Y] = (-1) \cdot P(Y = -1) + 0 \cdot P(Y = 0) + 1 \cdot P(Y = 1) = -\frac{2}{8} + \frac{4}{8} = \frac{1}{4}$$
$$E[Y^2] = (-1)^2 \cdot P(Y = -1) + 0^2 \cdot P(Y = 0) + 1^2 \cdot P(Y = 1) = \frac{2}{8} + \frac{4}{8} = \frac{3}{4}$$

Hence,

$$\mathrm{Var}(Y) = E[Y^2] - (E[Y])^2 = \frac{3}{4} - \frac{1}{16} = \frac{11}{16}.$$

(d) (3 points) Find $P(Y < X)$.

Solution: We have

$$P(Y < X) = 1 - P(Y \geq X) = 1 - P(1, 1) = 1 - \frac{2}{8} = \frac{3}{4}.$$

3. (8 points) Let $X \sim \text{Unif}\,[0,5]$. Find the probability density function of the random variable $Y = X^2$.

Solution: We will first find the cumulative distribution function (cdf) $F_Y(t)$ of Y.

The range of X is $[0,5]$, so the range of $Y = X^2$ is $[0,25]$.

Then we have, using that $X \sim \text{Unif}\,[0,5]$ and so $P(X \in [0,a]) = \frac{a}{5}$ for any $a \in [0,5]$:

$$F_Y(t) = P(X^2 \le t,\ X \in [0,5]) = 0, \qquad \text{if } \ t < 0;$$

$$F_Y(t) = P(X^2 \le t,\ X \in [0,5]) = P(X \in [0,\sqrt{t}]) = \frac{\sqrt{t}}{5}, \qquad \text{if } \ 0 \le t < 25;$$

$$F_Y(t) = P(X^2 \le t,\ X \in [0,5]) = 1, \qquad \text{if } \ t > 25.$$

Note that $F_Y(t)$ is continuous (we need to check that only at the points $t = 0$ and $t = 25$), and $F'(t)$ exists everywhere except these two points. Then the probability density function $f_Y(t)$ of Y is obtained by differentiating $F_Y(t)$, except the points $t = 0$ and $t = 25$ where we define $f_Y(t)$ arbitrarily. Then we have

$$f_Y(t) = \begin{cases} 0, & t \le 0 \\ \frac{1}{10\sqrt{t}}, & 0 < t < 25 \\ 0, & t \ge 25. \end{cases}$$

4. (6 points) The owner of a certain website is studying the distribution of the number of visitors to the site. Every day, a million $(1,000,000)$ people independently decide whether to visit the site, with probability $p = 2 \cdot 10^{-6}$ of visiting. Give a good approximation for the probability of getting at least three visitors on a particular day.

Solution: Let $X \sim \text{Bin}(n, p)$ be the number of visitors, where $n = 10^6$. Since n is large, p is small, and np^2 is very small (< 0.1), Poisson(2) is a good approximation. This gives

$$P(X \geq 3) = 1 - P(X < 3) \approx 1 - e^{-2} - e^{-2} \cdot 2 - e^{-2} \cdot \frac{2^2}{2!} = 1 - 5e^{-2} \approx 0.3233,$$

5. Let random variables X and Y have joint pdf

$$\begin{cases} A(x,y) & \text{for } 0 \le x \le 4, 0 \le y \le 3, \\ 0 & \text{otherwise} \end{cases}$$

Your integrals in the answers in this problem should be of the form $\int_a^b \int_c^d \ldots$ with appropriate limits of integration a, b, c, d and an explicit function inside the integral.

(a) (4 points) Which conditions do we need to check to be sure that this is actual joint pdf?

Solution: $\int_0^4 \int_0^3 A(x,y)dydx = 1$ and $A(x,y) \ge 0$ for any $0 \le x \le 4$, $0 \le y \le 3$.

(b) (4 points) Write an integral expression for the $E[X^2Y]$.

Solution:

$$E[X^2Y] = \int_0^4 \int_0^3 x^2 y A(x,y)dydx.$$

(c) (4 points) Write an integral expression for the $P(X+Y<2)$.

Solution:

$$P(X+Y<2) = \int_0^2 \int_0^{2-x} A(x,y)dydx$$

6. Suppose random variable X has density function

$$f_X(x) = \begin{cases} \dfrac{2}{x^3}, & x \geq 1 \\ 0, & x < 1 \end{cases}$$

(a) (6 points) Find $P(2 \leq X \leq 3)$

Solution:

$$P(2 \leq X \leq 3) = \int_2^3 \frac{2}{x^3} dx = -\frac{1}{x^2}\bigg|_2^3 = -\frac{1}{9} + \frac{1}{4} = \frac{5}{36}.$$

(b) (6 points) Find $E[X]$

Solution:

$$E[X] = \int_1^\infty x \frac{2}{x^3} dx = \int_1^\infty \frac{2}{x^2} dx = -\frac{2}{x}\bigg|_1^\infty = 2.$$

(c) (6 points) Find $\mathrm{Var}[X]$

Solution:

$$E[X^2] = \int_1^\infty x^2 \frac{2}{x^3} dx = \int_1^\infty \frac{2}{x} dx.$$

Integral diverges, so $E[X^2] = +\infty$ and $\mathrm{Var}(X) = +\infty$.

7. (8 points) Our ice machines produce ice cubes, 3 or 4 ice cubes every second with equal probabilities. Approximate the probability that that in 1 minutes and 40 seconds we will have at least 360 ice cubes.

Solution: Let $T_i = 3 + X_i$ be the number of cubes produced in the ith second, $X_i \sim Ber(\frac{1}{2})$, and let $S_{100} = 300 + X_1 + \ldots + X_{100}$ be the total number of cubes produced during 100 seconds. Note that $S_{100} \sim 300 + Bin(100, \frac{1}{2})$, $100 \cdot \frac{1}{2} \cdot \frac{1}{2} > 10$, so we can use Normal approximation

$$P(S_{100} \geq 360) = P(X_1 + \ldots + X_{100} \geq 60) = P\left(\frac{X_1 + \ldots + X_{100} - 50}{\sqrt{100 \cdot \frac{1}{2} \cdot \frac{1}{2}}} \geq \frac{60 - 50}{5}\right)$$

$$\approx 1 - \Phi(2) = 0.0228.$$

First Name: _____ Last Name: _____

Table of values for $\Phi(x)$, the CDF of a standard normal random variable

	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998

SCRATCH PAPER - DO NOT REMOVE FROM YOUR EXAM.
SCRATCH WORK WILL NOT BE GRADED

Second Midterm Exam

Name: _____

Please circle your section number!

Section 1	Section 2	Section 3	Section 4
Chenxi Wu (TR 1:00 PM)	Niclas Technau (MWF 1:20 PM)	Ander Aguirre (TR 9:30 AM)	Sarah Tammen (MWF 12:05 PM)

- **There are 6 problems on the exam, some of them have multiple parts.**
- Read the problems carefully and budget your time wisely.
- If you are running out of time, please at least say *how* you would solve a given problem, preferably using notation, even if you do not have the time to do so. This may earn you partial credit.
- No calculators or other electronic devices, please. **Turn off your phone.**
- You do not have to carry out complicated numerical computations, but you should simplify your answer if it is possible with reasonable effort. In particular, geometric series must be evaluated.
- Please present your solutions in a clear manner. **Justify your steps.** Numerical answers without explanation cannot get credit. Cross out the writing that you do not wish to be graded on.
- You can use the empty pages for additional space for writing if necessary.

Question	Points	Score
1	18	
2	10	
3	20	
4	19	
5	15	
6	18	
Total:	100	

1. In any parts of the following problem for which you deem it relevant, you may leave answers in terms of the function Φ (the CDF of a standard normal random variable) or its inverse.

   (a) (5 points) The dean of a certain university wants to estimate the proportion of students that are left-handed. He plans to send a survey to a random sample of the student body. If the dean wants to obtain a 95% confidence interval for the proportion of students that are left-handed and he wants an interval of width ε, then how many students should he survey? Give an answer of the form $n \geq f(\varepsilon)$.

   (b) (5 points) Suppose the dean's survey finds that the proportion of students that are left-handed is $p = 0.10$. Suppose that Professor Bob is teaching a 160 person Calculus class in a room with 20 left-handed desks. Use an appropriate approximation to estimate the probability that there will be more than 20 left-handed students in the class. (Assume that the hand preferences of different students are independent.)

(c) (5 points) Suppose that 0.1% (i.e. 1/1000) of the students in the university are ambidextrous. Suppose that the university admits a freshman class of $10,000$ students. Use an appropriate approximation to estimate the probability that exactly two of the new freshmen are ambidextrous.

(d) (3 points) We sample 40 students with replacement from the student body. Let X_1 be the number of times we choose a left-handed student, let X_2 be the number of times we choose an ambidextrous student, and let X_3 be the number of times we choose a right-handed student. Under the assumptions of parts (b)-(c), what distribution does the vector (X_1, X_2, X_3) have? (For full credit, you must specify all parameters.)

2. (10 points) Let X be a random variable with mean -4, and variance 16.
Find numbers $a, b > 0$ so that the random variable $Y = aX + b$ has mean 0 and variance 1.

3. Let $X \sim \text{Unif}([-1, 3])$. Compute the probability density functions of two the random variables:

(a) (10 points) $Y = X^3$

(b) (10 points) $Y = |X|$

4. In this problem we consider the moment generating function of a discrete random variable X.

(a) (11 points) Suppose the moment generating function of X is

$$M_X(t) = \frac{1}{8} + \frac{3}{8}e^t + \frac{3}{8}e^{2t} + \frac{1}{8}e^{3t}.$$

Derive the probability mass function of X.

(b) (7 points) Use the derivatives of M_X to find the variance $\mathrm{Var}(X)$.

(c) (1 point) Can you identify the distribution of X with parameters?

5. Let (X, Y) be uniformly distributed on the triangle with vertices $(0, 0)$, $(1, 1)$, and $(1, 2)$.

(a) (12 points) Find the marginals of the coordinates f_Y and f_X.

(b) (3 points) Are the two coordinates independent or not?

6. Let $\lambda > 0$. Let $X \sim \text{Exp}(\lambda)$ and $Y \sim \text{Exp}(2\lambda)$ be independent exponentials.

(a) (9 points) Compute the probability $P(X > 2Y)$. Write the answer in terms of λ.

(b) (9 points) Let $V = \min(X, Y)$. Derive a formula for the probability density function f_V of V.

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List of named distributions

$X \sim \text{Bernoulli}(p)$, PMF: $p_X(0) = 1 - p, \quad p_X(1) = p$

$X \sim \text{Binomial}(n, p)$, PMF: $p_X(k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad 0 \le k \le n$

$X \sim \text{Geometric}(p)$, PMF: $p_X(k) = p(1-p)^{k-1}, \quad k = 1, 2, \ldots$

$X \sim \text{Poisson}(\lambda)$, PMF: $p_X(k) = \frac{\lambda^k}{k!} e^{-\lambda}, \quad k = 0, 1, 2, \ldots$

$X \sim \text{Hypergeometric}(N, N_A, n)$, PMF: $p_X(k) = \frac{\binom{N_A}{k}\binom{N-N_A}{n-k}}{\binom{N}{n}}, \quad 0 \le k \le n$

$X \sim \text{Uniform}(a, b)$, PDF: $f_X(x) = \begin{cases} \frac{1}{b-a} & \text{for } x \in [a, b], \\ 0 & \text{otherwise} \end{cases}$

$X \sim \text{Normal}(\mu, \sigma^2)$, PDF: $f_X(t) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(t-\mu)^2}{2\sigma^2}}$

$X \sim \text{Exponential}(\lambda)$, PDF: $f_X(t) = \begin{cases} \lambda e^{-\lambda x} & \text{for } x > 0, \\ 0 & \text{otherwise} \end{cases}$

$X \sim \text{Gamma}(r, \lambda)$, PDF: $f_X(t) = \begin{cases} \frac{1}{\Gamma(r)} \lambda^r x^{r-1} e^{-\lambda x} & \text{for } x > 0, \\ 0 & \text{otherwise} \end{cases}$

$(X_1, \ldots, X_k) \sim \text{Multinomial}(n, k, p_1, \ldots, p_k)$,
joint PMF: $p(a_1, \ldots, a_k) = \binom{n}{a_1, \ldots, a_k} p_1^{a_1} \cdots p_k^{a_k}$ for $a_1, \ldots, a_k \ge 0$ with $a_1 + \cdots + a_k = n$.

The following table shows approximate values of the cumulative distribution function of the standard normal distribution.

x	$\Phi(x)$
-2.0	0.022
-1.9	0.028
-1.8	0.035
-1.7	0.044
-1.6	0.054
-1.5	0.066
-1.4	0.080
-1.3	0.096
-1.2	0.115
-1.1	0.135
-1.0	0.158
-0.9	0.184
-0.8	0.211
-0.7	0.242
-0.6	0.274
-0.5	0.308
-0.4	0.344
-0.3	0.382
-0.2	0.420
-0.1	0.460
0.0	0.500
0.1	0.539
0.2	0.579
0.3	0.617
0.4	0.655
0.5	0.691
0.6	0.725
0.7	0.758
0.8	0.788
0.9	0.815
1.0	0.841
1.1	0.864
1.2	0.884
1.3	0.903
1.4	0.919
1.5	0.933
1.6	0.945
1.7	0.955
1.8	0.964
1.9	0.971
2.0	0.977

Second Midterm Exam - Solutions

Name: _____

Please circle your section number!

Section 1	Section 2	Section 3	Section 4
Chenxi Wu (TR 1:00 PM)	Niclas Technau (MWF 1:20 PM)	Ander Aguirre (TR 9:30 AM)	Sarah Tammen (MWF 12:05 PM)

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1. In any parts of the following problem for which you deem it relevant, you may leave answers in terms of the function Φ (the CDF of a standard normal random variable) or its inverse.

   (a) (5 points) The dean of a certain university wants to estimate the proportion of students that are left-handed. He plans to send a survey to a random sample of the student body. If the dean wants to obtain a 95% confidence interval for the proportion of students that are left-handed and he wants an interval of width ε, then how many students should he survey? Give an answer of the form $n \geq f(\varepsilon)$.

   Solution: Let p be the true proportion of the student body that is left-handed. If the dean wants a confidence interval of width ε for p, then he wants a margin of error $\frac{\varepsilon}{2}$. That is, if we let S_n be the number of students in a sample of n that are left-handed, he wants to guarantee that

$$P\left(\left|\frac{S_n}{n} - p\right| < \frac{\varepsilon}{2}\right) \geq 0.95.$$

   For this, it is sufficient that

$$0.95 \leq 2\Phi\left(2\left(\frac{\varepsilon}{2}\right)\sqrt{n}\right) - 1 = 2\Phi(\varepsilon\sqrt{n}) - 1.$$

   We solve for n to give the condition

$$n \geq \left(\frac{\Phi^{-1}(0.975)}{\varepsilon}\right)^2.$$

   Note: if you interpreted the question to mean that the margin of error should be ε, then we will give full credit. Under this assumption, you should end up with the inequality

$$n \geq \left(\frac{\Phi^{-1}(0.975)}{2\varepsilon}\right)^2.$$

   (b) (5 points) Suppose the dean's survey finds that the proportion of students that are left-handed is $p = 0.10$. Suppose that Professor Bob is teaching a 160 person Calculus class in a room with 20 left-handed desks. Use an appropriate approximation to estimate the probability that there will be more than 20 left-handed students in the class. (Assume that the hand preferences of different students are independent.)

   Solution: Let X_L be the number of left-handed students in Professor Bob's calculus class. If we assume that the hand preferences of different students are independent, then $X \sim \text{Bin}(n, p)$, where $n = 160$ and $p = \frac{1}{10}$. We want to estimate $P(X_L > 20)$. We plan to use the normal approximation for the binomial distribution; for this we must check that $np(1-p) \geq 10$. We see that

$$np(1-p) = 160\left(\frac{1}{10}\right)\left(\frac{9}{10}\right) = \frac{16 \times 9}{10} = \frac{12 \times 12}{10} > \frac{10 \times 12}{10} = 12 > 10.$$

   Thus, we can approximate $\frac{X-16}{12/\sqrt{10}}$ with a standard normal random variable. We have that

$$P(X > 20) = P\left(\frac{X-16}{12/\sqrt{10}} > \frac{20-16}{12/\sqrt{10}}\right) \approx 1 - \Phi\left(\frac{4}{12/\sqrt{10}}\right) = 1 - \Phi\left(\frac{\sqrt{10}}{3}\right).$$

(c) (5 points) Suppose that 0.1% (i.e. 1/1000) of the students in the university are ambidextrous. Suppose that the university admits a freshman class of $10,000$ students. Use an appropriate approximation to estimate the probability that exactly two of the new freshmen are ambidextrous.

Solution: Let X_A be the number of freshmen that are ambidextrous. Assuming that the hand preferences of different students are independent, we have that $X_A \sim \text{Binom}(m, q)$ with $m = 10000$ and $q = \frac{1}{1000}$. We want to compute $P(X_A = 2)$. We compute mq^2:

$$mq^2 = 10000\left(\frac{1}{1000}\right)^2 = \frac{10}{1000} = \frac{1}{100}.$$

Since $mq^2 \ll 1$, a Poisson approximation is appropriate. Specifically, if we let $Y \sim \text{Poisson}(mq)$, then we have

$$P(X_A = 2) \approx P(Y = 2) = e^{-10}\left(\frac{10^2}{2!}\right) = \frac{50}{e^{10}}.$$

(d) (3 points) We sample 40 students with replacement from the student body. Let X_1 be the number of times we choose a left-handed student, let X_2 be the number of times we choose an ambidextrous student, and let X_3 be the number of times we choose a right-handed student. Under the assumptions of parts (b)-(c), what distribution does the vector (X_1, X_2, X_3) have? (For full credit, you must specify all parameters.)

Solution: Letting p, q be as in parts (b)-(c), we have that

$$1 - p - q = \frac{899}{1000}.$$

Thus, the random vector (X_1, X_2, X_3) has the distrubution $\text{Multinom}\left(40, 3, \frac{1}{10}, \frac{1}{1000}, \frac{899}{1000}\right)$.

As an alternative to giving the name of the distribution, students may give the probability mass function: for any nonnegative integers k_1, k_2, k_3 with $k_1 + k_2 + k_3 = 40$, we have that

$$P(X_1 = k_1, X_2 = k_2, X_3 = k_3) = \binom{40}{k_1, k_2, k_3}\left(\frac{1}{10}\right)^{k_1}\left(\frac{1}{1000}\right)^{k_2}\left(\frac{899}{1000}\right)^{k_3}.$$

2. (10 points) Let X be a random variable with mean -4, and variance 16.
Find numbers $a, b > 0$ so that the random variable $Y = aX + b$ has mean 0 and variance 1.

Solution: The random variable $Z = X - (-4) = X + 4$ has mean 0. Note $Y = aZ$ has variance $\mathrm{Var}(Z) = a^2\mathrm{Var}(X) = a^2 16$. Choosing $a = 1/4$ ensures Y has variance one. Thus $b = 4a = 1$.

3. Let $X \sim \text{Unif}([-1,3])$. Compute the probability density functions of two the random variables:

(a) (10 points) $Y = X^3$

Solution: The function $g : x \mapsto x^3$ is strictly increasing, in particular on $[-1,3]$. Hence, the probability density functions f_Y of Y and f_X of X are related by the change of variables formula

$$f_Y(s) = f_X(g^{-1}(s))\left|\frac{\mathrm{d}}{\mathrm{d}s}g^{-1}(s)\right|$$

where s is in the range of Y given by $[g(-1), g(3)] = [-1, 27]$. Presently, $g^{-1}(s) = s^{1/3}$. Its derivative exists and equals $\frac{1}{3}s^{-2/3}$ as long as $s \neq 0$. If $s \in [-1, 27]$, then $f_X(g^{-1}(s)) = 1/4$. Consequently,

$$f_Y(s) = \frac{1}{12|s^{2/3}|}$$

if $s \in (-1, 27) \setminus \{0\}$, and $f_Y(s) = 0$ for any other s.

The lack of differentiability of the cumulative distribution function $F_Y(s) = F_X(s^{1/3})$ at $s = 0$ is irrelevant. The density functions f_Y can be assigned any value there. We chose put the f_Y to equal zero there for the ease of exposition.

(b) (10 points) $Y = |X|$

Solution: The range of Y is $[0,3]$. We compute its cumulative distribution function

$$F_Y(s) = P(Y \leq s) = P(X \in [-s, s]).$$

If $s \in [0,1]$, then $F_Y(s) = \int_{-s}^{s} \frac{1}{4} = s/2$. If $s \in (1,3]$, then $F_Y(s) = \int_{-1}^{s} \frac{1}{4} = s/4 + 1/4$. Differentiating yields that a probability density function f_Y of Y is

$$f_Y(s) = F_Y'(s) = \begin{cases} 1/2, & \text{if } 0 < s < 1, \\ 1/4, & \text{if } 1 \leq s \leq 3, \\ 0, & \text{otherwise.} \end{cases}$$

4. In this problem we consider the moment generating function of a discrete random variable X.

(a) (11 points) Suppose the moment generating function of X is

$$M_X(t) = \frac{1}{8} + \frac{3}{8}e^t + \frac{3}{8}e^{2t} + \frac{1}{8}e^{3t}.$$

Derive the probability mass function of X.

Solution: Reading off from the MGF we obtain $P(X = 0) = \frac{1}{8}, P(X = 1) = \frac{3}{8}, P(X = 2) = \frac{3}{8}, P(X = 3) = \frac{1}{8}$.

(b) (7 points) Use the derivatives of M_X to find the variance $\mathrm{Var}(X)$.

Solution: We must differntiatie. For the first moment we have:

$$\mathbb{E}X = M'(0) = \frac{3}{8} + 2 \cdot \frac{3}{8} + 3 \cdot \frac{1}{8} = \frac{3}{2}$$

For the second moment we have:

$$\mathbb{E}X^2 = M''(0) = \frac{3}{8} + 2^2 \cdot \frac{3}{8} + 3^2 \cdot \frac{1}{8} = 3$$

And hence the variance is

$$\mathrm{var}(X) = \mathbb{E}X^2 - (\mathbb{E}X)^2 = \frac{3}{4}$$

(c) (1 point) Can you identify the distribution of X with parameters?

Solution: $\mathrm{Bin}(3, \frac{1}{2})$

5. Let (X, Y) be uniformly distributed on the triangle with vertices $(0,0)$, $(1,1)$, and $(1,2)$.

(a) (12 points) Find the marginals of the coordinates f_Y and f_X.

Solution: The area of the triangle is $1/2$, so the joint pdf is

$$f(x,y) = \begin{cases} 2 & 0 \le x \le 1, x \le y \le 2x \\ 0 & \text{otherwise} \end{cases}$$

The marginal pdfs are

$$f_X(x) = \int_{-\infty}^{\infty} f(x,y)dy = \begin{cases} \int_x^{2x} 2dy & 0 \le x \le 1 \\ 0 & \text{otherwise} \end{cases}$$

$$= \begin{cases} 2x & 0 \le x \le 1 \\ 0 & \text{otherwise} \end{cases}$$

$$f_Y(y) = \int_{-\infty}^{\infty} f(x,y)dx = \begin{cases} \int_{x/2}^{x} 2dx & 0 \le y \le 1 \\ \int_{x/2}^{1} 2dx & 1 \le y \le 2 \\ 0 & y < 0 \text{ or } y > 2 \end{cases}$$

$$= \begin{cases} y & 0 \le y \le 1 \\ 2-y & 1 < y \le 2 \\ 0 & y < 0 \text{ or } y > 2 \end{cases}$$

(b) (3 points) Are the two coordinates independent or not?

Solution: They are not independent. Because $P(0 < X < 1/2) = 1/4$, $P(Y > 1) = 1/2$, but $P(0 < X < 1/2, Y > 1) = 0$.

6. Let $\lambda > 0$. Let $X \sim \text{Exp}(\lambda)$ and $Y \sim \text{Exp}(2\lambda)$ be independent exponentials.

(a) (9 points) Compute the probability $P(X > 2Y)$. Write the answer in terms of λ.

Solution: Because X and Y are independent, their joint pdf is

$$f(x,y) = f_X(x)f_Y(y) = \begin{cases} 2\lambda^2 e^{-\lambda x - 2\lambda y} & x > 0, y > 0 \\ 0 & \text{otherwise} \end{cases}$$

So

$$P(X > 2Y) = \int_{x>2y} f(x,y)dxdy = \int_{x>2y>0} 2\lambda^2 e^{-\lambda x - 2\lambda y} dxdy$$

$$= \int_0^\infty \left(\int_{2y}^\infty \lambda e^{-\lambda x} dx \right) 2\lambda e^{-2\lambda y} dy = \int_0^\infty e^{-2\lambda y} 2\lambda e^{-2\lambda y} dy = \frac{2\lambda}{4\lambda} = 1/2$$

Alternatively, we see that $2Y \sim Exp(\lambda)$, so X and $2Y$ are i.i.d, hence $P(X > 2Y) = P(X < 2Y)$. And because they are jointly continuous, $P(X = 2Y) = 0$, hence $P(X > 2Y) = 1/2$.

(b) (9 points) Let $V = \min(X, Y)$. Derive a formula for the probability density function f_V of V.

Solution: Note that we have the following equality of events

$$\{V > v\} = \{X > v\} \cap \{Y > v\}.$$

Then by independence

$$\mathbb{P}(V > v) = \mathbb{P}(X > v) \cdot \mathbb{P}(Y > v) = e^{-\lambda v} e^{-2\lambda v} = e^{-3\lambda v}.$$

Therefore, the CDF and the PDF are given by

$$F_V(v) = 1 - e^{-3\lambda v} \qquad \Longrightarrow \qquad f_V(v) = 3e^{-3\lambda v}$$

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List of named distributions

$X \sim \text{Bernoulli}(p)$, PMF: $p_X(0) = 1 - p, \quad p_X(1) = p$

$X \sim \text{Binomial}(n, p)$, PMF: $p_X(k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad 0 \le k \le n$

$X \sim \text{Geometric}(p)$, PMF: $p_X(k) = p(1-p)^{k-1}, \quad k = 1, 2, \ldots$

$X \sim \text{Poisson}(\lambda)$, PMF: $p_X(k) = \frac{\lambda^k}{k!} e^{-\lambda}, \quad k = 0, 1, 2, \ldots$

$X \sim \text{Hypergeometric}(N, N_A, n)$, PMF: $p_X(k) = \frac{\binom{N_A}{k}\binom{N-N_A}{n-k}}{\binom{N}{n}}, \quad 0 \le k \le n$

$X \sim \text{Uniform}(a, b)$, PDF: $f_X(x) = \begin{cases} \frac{1}{b-a} & \text{for } x \in [a, b], \\ 0 & \text{otherwise} \end{cases}$

$X \sim \text{Normal}(\mu, \sigma^2)$, PDF: $f_X(t) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(t-\mu)^2}{2\sigma^2}}$

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$X \sim \text{Gamma}(r, \lambda)$, PDF: $f_X(t) = \begin{cases} \frac{1}{\Gamma(r)} \lambda^r x^{r-1} e^{-\lambda x} & \text{for } x > 0, \\ 0 & \text{otherwise} \end{cases}$

$(X_1, \ldots, X_k) \sim \text{Multinomial}(n, k, p_1, \ldots, p_k)$,
joint PMF: $p(a_1, \ldots, a_k) = \binom{n}{a_1, \ldots, a_k} p_1^{a_1} \cdots p_k^{a_k}$ for $a_1, \ldots, a_k \ge 0$ with $a_1 + \cdots + a_k = n$.

The following table shows approximate values of the cumulative distribution function of the standard normal distribution.

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